

Kushal Mamillapalli

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EDUCATION

New York University

Brooklyn, NY

B.S. in Computer Science, Minor in Mathematics | GPA: 3.9/4.0, Dean's List

Sep. 2023 - May 2026

- Coursework: Data Structures, Algorithms, Operating Systems, Databases, Machine Learning, Data Science.

TECHNICAL SKILLS

Languages: Scala, Java, Python, C++, Bash, sbt, Maven

Data & Streaming: Apache Beam/Scio, Google Cloud Dataflow, BigQuery (SQL), Bigtable, Pub/Sub, Protobuf, Avro

ML & Vision: ROS 2, TensorRT, DeepStream, CUDA/NVMM, PyTorch, YOLOv8, scikit-learn, Prophet, Optuna

Cloud Infrastructure: Google Cloud IAM, Kubernetes, GitHub Actions/CI

Certifications: AWS Certified Machine Learning Engineer - Associate (May 2026)

EXPERIENCE

Spotify

Jun. 2026 – Aug. 2026

Data Engineering Intern

New York City, NY

- Built a Claude/MCP deprecation skill for Spotify across 10+ repos with reader checks, SQL parity, and PR summaries, cutting manual time 70%.
- Decommissioned 2 legacy Dataflow pipelines and 3 Pub/Sub topics, reducing GCP spend by ~\$15K/year.
- Unblocked recommendation training with naming standards on petabyte-scale gRPC services, backward-compatible for 10+ teams on legacy versions (16 renames).
- Fixed a validation job computing against all data models instead of the relevant subset, cutting compute cost by 4x–100x depending on pipeline.

NYU Enterprise Data Management

Apr. 2025 – Present

Technical Systems Engineering Intern (Part-Time)

New York City, NY

- Reduced runtime 40% for NYU Financial Aid's ETL pipeline by optimizing SQL/Pandas and migrating 100GB+ inputs to columnar Parquet, reducing storage and improving read times 54%.
- Developed a 600K-row SARIMA/Prophet forecasting engine with 4% MAPE, visualizing departmental trends on an Oracle Analytics Cloud dashboard.
- Established the team's first GitHub version-control workflow and removed 100+ legacy files and secrets from the server, improving security and cutting debugging time 75%.

NYU ARC Robotics

Jan. 2024 – Jun. 2026

Computer Vision Lead

New York City, NY

- Engineered the NYU ARC Robotics League robot's C++ ROS 2 auto-aim system with Basler/RealSense, YOLO, and TensorRT/DeepStream.
- Reduced latency 70% (40ms to 12ms) and raised throughput 2.3x (~67 to 156 FPS) with CUDA/NVMM.
- Boosted armor-targeting F1 15% with tracking, 3D PnP/ballistics, and STM32 UART commands.
- Containerized Jetson vision stack with Docker/systemd, health checks, model selection, and tooling.

PROJECTS

NBAnomaly | [GitHub](#)

Dec. 2025 – Present

- Developed a general-use NBA platform to surface player under- and overperformance across 10+ seasons with FastAPI on AWS EC2 and resilient ingestion.
- Validated 78% backtest precision with an Isolation Forest over 15+ engineered features.
- Lowered LLM report latency from ~4.5s to 2.5ms with a cache-aside PostgreSQL layer.

AutoCPT (HackNYU 2025 Winner) | [GitHub](#)

Feb. 2025

- Created a live-visit assistant that turns clinician-patient voice conversations into suggested CPT codes, saving clinicians 16+ hours/week with Whisper, LLaMA-3, and Groq.
- Enabled <500ms live responses with async Flask, retry/audit logic, and React YOLOv8 overlays.